

Question Number	Answer	Additional guidance	Mark
3(a)(i)	A calculation in which: • calculation of volume of sphere (1) • calculation of volume of hemisphere to nearest whole number (1)	<u>Example of calculation:</u> answer between 65416.67 to 65476.19 whole number answer between 32708 to 32738 Correct answer scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	An explanation that makes reference to two of the following: • (because) calabash fruits are renewable / more can be grown / will not run out / available to future generations (1) • biodegradable / can be broken down by decomposers (1) • carbon neutral (1)	ignore produces less greenhouse gases	(2)

Question Number	Answer	Mark
3(b)(i)	The only correct answer is B one <i>A is not correct because they provide support to the plant</i> <i>B is not correct because they are not involved in transporting substances</i> <i>C is not correct because they are not involved in transporting substances</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that makes reference to four of the following: Similarities: • both (fibres) contain cellulose (in the cell wall) (1) • both have tubular structures (1) • both do not contain a nucleus (1) Differences (max 3): • phloem (sieve tubes) have {sieve plates / (perforated) end walls} whereas xylem (vessels) have no {end walls / sieve plates} (1) • phloem (sieve tubes) {contain cytoplasm / are not hollow} whereas xylem (vessels) {do not contain cytoplasm / are hollow} (1) • phloem (sieve tubes) contain no {lignin / secondary thickening} whereas xylem contain {lignin / secondary thickening} (1) • phloem have plasmodesmata whereas xylem have pits (1)	ACCEPT piecing together from adjacent sentences ACCEPT both have a cell wall ignore organelles ACCEPT phloem (sieve tubes) contain living cells whereas xylem (vessels) contain dead cells ACCEPT phloem cell walls are thinner than xylem cell walls / converse ACCEPT xylem have pits whereas phloem do not / phloem have plasmodesmata whereas xylem do not	(4)